AWS Infrastructure Troubleshooting Report

Sample Work — Demonstration Environment

Project Overview

This report documents AWS infrastructure troubleshooting and configuration work performed in an AWS-hosted demonstration environment.

The environment was designed to support Windows-based infrastructure, shared storage, Active Directory, and application delivery. Troubleshooting involved AWS compute, networking, storage, and integration with Windows Server infrastructure.

The work demonstrates a structured approach to identifying infrastructure issues, examining the AWS environment, validating configuration, and determining the appropriate resolution.

AWS Environment

The demonstration environment included:

- Amazon EC2
- Amazon VPC
- Amazon FSx for Windows File Server
- Windows Server 2022
- Active Directory Domain Services
- DNS
- Parallels RAS application delivery
- Windows-based virtual desktop infrastructure
- MSIX/VHDX application packages

Challenge

The environment required multiple AWS services and Windows infrastructure components to operate together.

Troubleshooting required validating that EC2 instances could communicate correctly within the VPC, Windows Server systems could communicate with Active Directory and DNS, shared storage was accessible, and application delivery components could use the underlying infrastructure.

Troubleshooting Approach

1. Verify EC2 Infrastructure

EC2 instances were reviewed to verify instance status, configuration, connectivity, and their role within the overall infrastructure.

2. Verify VPC Configuration

The VPC environment was reviewed to verify networking components and connectivity between the required infrastructure resources.

3. Verify Storage Integration

Amazon FSx for Windows File Server was reviewed as part of the shared storage environment supporting Windows infrastructure and application delivery.

4. Verify Windows Integration

Windows Server systems were evaluated to confirm domain connectivity, DNS functionality, authentication, and communication with AWS-hosted infrastructure.

5. Isolate the Problem

When an application or service did not operate as expected, the investigation separated AWS infrastructure issues from Windows Server, Active Directory, storage, and application-delivery configuration issues.

This helped identify whether the problem originated within AWS or from a dependent component.

AWS Services Used

The environment included hands-on work with:

- Amazon EC2
- Amazon VPC
- Amazon FSx for Windows File Server
- AWS networking components
- Windows Server infrastructure hosted in AWS

Windows Infrastructure Integration

The AWS environment was not treated as an isolated cloud environment. AWS infrastructure was integrated with Windows Server services and application delivery components.

Active Directory and DNS provided centralized Windows infrastructure services, while Amazon FSx provided shared Windows file storage. Parallels RAS was used for application delivery within the environment.

This required troubleshooting across both the AWS infrastructure layer and the Windows infrastructure layer.

Findings

The troubleshooting process demonstrated the importance of examining the complete infrastructure rather than assuming that an application or service problem originates from AWS itself.

EC2, VPC networking, Windows Server, Active Directory, DNS, shared storage, and application delivery components all had to be considered when isolating infrastructure issues.

Resolution

The AWS environment was configured to provide the required compute, networking, and storage infrastructure for the Windows-based environment.

AWS-hosted Windows systems were integrated with Active Directory, DNS, Amazon FSx, and application delivery components to support the overall infrastructure.

Result

The completed environment provided a functional AWS-hosted Windows infrastructure platform capable of supporting centralized identity services, shared storage, and application delivery.

The troubleshooting process provided a repeatable method for investigating AWS infrastructure issues by working from the underlying cloud resources through networking, storage, Windows integration, and application dependencies.

Key Takeaway

EC2 → VPC → Networking → Storage → Windows Infrastructure → Application Dependencies → Resolution

This layered approach helps identify the actual source of AWS infrastructure problems instead of applying changes without first determining the root cause.

Demonstration Environment Notice

This report describes work performed in a demonstration/lab environment and is provided as a sample of technical troubleshooting methodology and AWS infrastructure experience.